Scaling Solutions to the Temperature Field of Turbulent Boundary Layer Flows

Geoffrey Zheng

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Department of Physics
Princeton University
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Adviser: Professor Marcus N. Hultmark, Department of Mechanical and Aerospace Engineering
Second Reader: Professor Lyman A. Page, Jr., Department of Physics

This paper represents my work in accordance with University regulations.

/s/ Geoffrey Zheng

Abstract

The study of turbulence in fluid mechanics is motivated by our need to understand how quantities like momentum and heat are transported and dispersed throughout a flow field. Experimental measurements of such statistics in turbulent flows is limited and there exist several theories on the mean scalar behavior of the temperature field in a turbulent boundary layer flow. In this junior paper, I first provide a theoretical background on momentum and heat transport in turbulent boundary layers. I then take measurements of the mean value and variance of the temperature field in the turbulent boundary layer of a heated, re-circulating water tunnel using a state-of-the-art cold-wire resistance thermometer probe. The data gathered is analyzed to determine the effectiveness of a proposed temperature variance scaling parameter in collapsing the dynamics of flows with friction Reynolds number (Re_τ) of 500 to 1200. The results from this paper are used to provide a preliminary measure of the validity of current theories on temperature variance scaling parameters in turbulent boundary layer flows.

Contents

1 Introduction **1**
1.1 Turbulence 1
1.2 Dimensional Analysis and Scaling 3

2 Turbulent Boundary Layer Theory **5**
2.1 Momentum Transport in Turbulent Boundary Layer Flow 6
2.2 Heat Transport in Turbulent Boundary Layer Flow 8
2.3 Thermal Boundary Layer Scaling Parameters 10

3 Experimental Setup and Data Collection **14**
3.1 The Nanoscale Cold-Wire Resistance Thermometer Probe 14
3.2 The Re-circulating Water Tunnel and Heating Elements 14

4 Results **16**

5 Conclusions and Future Work **20**

A Appendix: Similarity Solution Method **22**

References **24**

Acknowledgements **25**

1 Introduction

A fluid is broadly defined as a substance that continually deforms under the application of shear stresses [1]. Fluid mechanics is the study of the behavior of fluids under the action of applied forces. One characteristic of fluid mechanics is the use of the continuum hypothesis, which states that the macroscopic behavior of fluids is the same as if the fluid were perfectly continuous in structure [2]. The hypothesis ignores the physical reality that macroscopic behavior comes from complex atomic and molecular interactions and remains valid as long as the characteristic length of the fluid system remains much larger than inter-atomic distances. Because of the continuum hypothesis, properties of the fluid can be described in terms of fields, such as the velocity field or the temperature field.

Closely related to fluid mechanics is the topic of transport theory. Transport theory concerns the study of non-equilibrium situations where quantities such as mass, momentum or heat are transported from one place to another, although locally the fluid is assumed to be in thermodynamical equilibrium [3]. The behavior of heat transfer in flow phenomena is of great physical interest because of its fundamental applications in science and engineering. Mechanical engineers are interested in heat transfer because of it governs the behavior of heat exchanger devices such as air conditioners and space heaters. Physicists are interested in heat transfer because of its application to laser cooling of atoms to near absolute zero temperatures. Because turbulent flows are the norm rather than the exception in nature, it is essential that we understand the behavior of turbulent momentum and heat transfer.

1.1 Turbulence

Turbulent flows are abundant throughout nature; they occur in situations like the rising smoke from a candle or the growing boundary layers on the wing of an airplane. Some defining features of turbulent flows include [4]:

- **Irregularity**: Turbulent flows are chaotic, with large 3-D unsteady vortical motions.
- **Rapid Transport**: Turbulent flows are characterized by efficient mixing of momentum and heat.
- **Large Range of Spatio-Temporal Scales**: Turbulent flows encompass an enormous range of flow size and time scales. The largest turbulent structures are sev-

eral orders of magnitude larger than the smallest structures, with the range being parameterized by the Reynolds number.

- **Large Reynolds Number**: Turbulent flows often originate from an instability of laminar flows if the Reynolds number grows too large, which arises due to the interaction of a viscous term ($\mu\nabla^2\vec{u}$) and nonlinear inertia term ($\vec{u}\cdot\nabla\vec{u}$) in the Navier-Stokes Equation (NSE).

Turbulent flows are excellent at transporting momentum and heat. In fact, the rate of transfer of these quantities due to turbulence is orders of magnitude greater than the rate of the transfer due to molecular diffusion. This is shown quantitatively in [4], and the argument is summarized as follows: the one-dimensional heat diffusion equation is given by $\frac{\partial\theta}{\partial t} = \gamma\frac{\partial^2\theta}{\partial x^2}$, where θ denotes temperature and γ denotes the thermal diffusivity which is taken to be constant. By non-dimensionalization, the characteristic time constant for molecular diffusion of heat, denoted t_d, is $t_d = \frac{L^2}{\gamma}$, where L is the characteristic length scale of the system. On the other hand, turbulence transports heat according to the timescale $t_t \sim \frac{L}{u}$, where u is the rms amplitude of velocity fluctuations. Thus $\frac{t_d}{t_t} \sim \frac{uL}{\gamma}$. For gases and most common liquids, $\nu/\gamma < 10$, so up to an order of magnitude, γ is replaced with ν ($\nu \equiv \mu/\rho$ is defined as the kinematic viscosity). This gives:

$$\frac{t_d}{t_t} \sim \frac{uL}{\nu} = \mathrm{Re}, \tag{1}$$

where Re is the Reynolds number. Since turbulent flows have high Re values, $t_t \ll t_d$. Thus turbulent time scales are much more rapid than molecular diffusion time scales, supporting the observation that turbulence is more effective than diffusion at transporting quantities.

At the same time, although most flows in nature are turbulent, there exist many problems with accurately analyzing them. Because turbulent flow encompasses such a large range of scales, it is difficult to conduct experiments to accurately measure various characteristics of the flow. Moreover, the statistical equations that govern turbulence suffer from the closure problem. This means that there always exists more unknown quantities than equations in any statistical description of turbulent behavior. To reduce the number of unknown quantities in order to match the number of equations, numerical models and estimates are made based on physical intuition. Finally, the nonlinearity of the NSE means it is not yet solved analytically. Therefore, approximation techniques such as dimensional analysis, similarity solutions, and asymptotic matching are needed

to study the behavior of turbulent flows.

1.2 Dimensional Analysis and Scaling

A powerful analytical tool used in turbulence theory is dimensional analysis, where a governing equation is non-dimensionalized to parameterize the flow in terms of particular combinations of the existing independent parameters. These new parameters can then be used in a similarity solution to the problem of interest. If a similarity solution indeed exists, the number of independent parameters is reduced and the solution will collapse, revealing a universal behavior of the system. An example of non-dimensionalization is given below to demonstrate this method. An example illustrating how the method of similarity solutions works is given in the appendix.

Consider a two-dimensional incompressible flow with the following conditions. The velocity field is given by $\vec{u} = (u_x, u_y, 0)$, $u_y \ll u_x$, $\dfrac{\partial u_x}{\partial x} \ll \dfrac{\partial u_x}{\partial y}$, the pressure gradient is in the x-direction, and gravity is in the y-direction and can hence be neglected. This setup is similar to the type of flow that is studied later on in the paper. A diagram of the flow is shown in Figure 1 on page 5. The governing equation of this flow is:

$$\rho\left(\frac{\partial u_x}{\partial t} + u_x\frac{\partial u_x}{\partial x} + u_y\frac{\partial u_x}{\partial y}\right) = -\frac{\partial p}{\partial x} + \mu\frac{\partial^2 u_x}{\partial y^2}. \tag{2}$$

Equation (2) is a simplified form of the NSE, which is a set of three non-linear, second order partial differential equations, one for each spatial direction. The flow is incompressible, so the continuity condition $\nabla \cdot \vec{u} = 0$ must be satisfied. ρ denotes fluid density, p denotes pressure, and μ denotes dynamic viscosity.

To rewrite equation (2) in non-dimensional form, the velocity, time, spatial derivative, and pressure are scaled by the following scaling parameters:

$$u_x^* = \frac{u_x}{u_c}, \quad u_y^* = \frac{u_y}{u_c}, \quad t^* = \frac{t}{t_c}, \quad \nabla^* = \ell\nabla, \quad p^* = \frac{p}{p_c}, \tag{3}$$

where the asterisk superscript denotes the non-dimensional version of the variable, ∇ denotes and ℓ is the characteristic length scale. Substituting these definitions into equation

(2) and using the chain rule gives:

$$\frac{\partial(u_c u_x^*)}{\partial(t_c t^*)} + \frac{u_c u_x^*}{\ell}\frac{\partial(u_c u_x^*)}{\partial x^*} + \frac{u_c u_y^*}{\ell}\frac{\partial(u_c u_x^*)}{\partial y^*} = -\frac{1}{\rho\ell}\frac{\partial(p_c p^*)}{\partial x^*} + \frac{\mu}{\rho\ell^2}\frac{\partial^2(u_c u_x^*)}{\partial y^{*2}} \tag{4}$$

$$\implies \left[\frac{\ell}{t_c u_c}\right]\frac{\partial u_x^*}{\partial t^*} + u_x^*\frac{\partial u_x^*}{\partial x^*} + u_y^*\frac{\partial u_x^*}{\partial y^*} = -\left[\frac{p_c}{\rho u_c^2}\right]\frac{\partial p^*}{\partial x^*} + \left[\frac{\mu}{\rho u_c \ell}\right]\frac{\partial^2 u_x^*}{\partial y^{*2}}. \tag{5}$$

Since equation (5) is non-dimensional, all terms are of order one. Therefore, we should impose that $\frac{\ell}{t_c u_c} = 1$ and $\frac{p_c}{\rho u_c^2} = 1$. Hence $t_c = \frac{\ell}{u_c}$ and $p_c = \rho u_c^2$.

The last square-bracketed term is important because it is multiplied by the highest-order derivative term which represents viscous stresses. This non-dimensional constant is known as the Reynolds number (Re):

$$\mathrm{Re} = \frac{\rho u \ell}{\mu}, \tag{6}$$

where ℓ is the characteristic length scale of the system. The Reynolds number, among other things, describes the ratio of inertial forces to viscous forces in the flow field.

Because the first two square-bracketed terms are set to be 1, the governing equation becomes:

$$\frac{\partial u_x^*}{\partial t^*} + u_x^*\frac{\partial u_x^*}{\partial x^*} + u_y^*\frac{\partial u_x^*}{\partial y^*} = -\frac{\partial p^*}{\partial x^*} + \frac{1}{\mathrm{Re}}\frac{\partial^2 u_x^*}{\partial y^{*2}}. \tag{7}$$

Comparing equation (2) with equation (7), the latter equation only contains one non-dimensional parameter, the Reynolds number. This parameter characterizes the flow in the sense that two flows with different physical properties, such as viscosities or characteristic lengths, will be governed by the same dynamics as long as they have the same Reynolds number. While turbulent flow is more complex and contains multiple regions governed by different parameters, the theme of this section still motivates the ultimate goal: to find the correct scaling parameters that will non-dimensionalize the governing equation in order to obtain a collapse of data that is governed by the same dynamics.

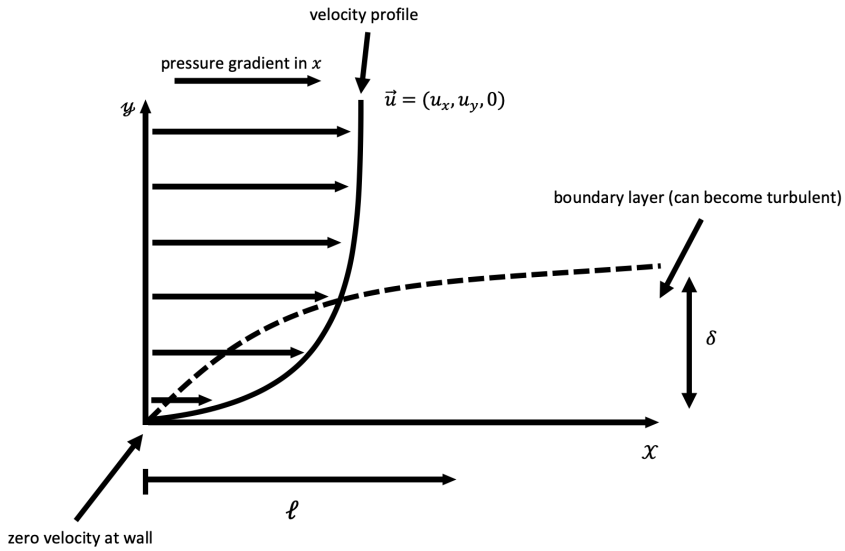


Figure 1: Diagram of a typical 2-D boundary layer flow (not to scale). The dotted line represents the upper boundary of the boundary layer. The solid curve represents the magnitude of the velocity field, which goes from zero at the wall to free-stream value far away from boundary layer. ℓ is the characteristic x-length and δ is the characteristic y-length.

2 Turbulent Boundary Layer Theory

When analyzing flow fields near a solid surface, viscous effects become important. The reason is that there arises a strong interaction between the molecules of the fluid and the molecules of the solid. From this boundary condition, the no-slip condition is obtained, which states that for a stationary surface, the velocity of the flow field in the region near the wall must reduce to zero [1]. Therefore there must exist steep velocity gradients in the region near the wall to bring the velocity of the flow field from 0 at the wall to free-stream values far away from the wall. The region where these strong gradients occur is called the boundary layer. Another boundary condition that must be satisfied by the flow is the kinematic boundary condition. The kinematic boundary condition states that the normal velocity of the fluid at the surface must equal the normal velocity of the surface, so there cannot be any flow through the surface [5].

The thickness of the boundary layer acts as a natural length scale in the wall-normal direction. A commonly used length scale derived from the thickness is called δ_{99}, and

is defined as the distance away from the wall at which the flow field velocity attains 99% of its free-stream value. Naturally then, a commonly used velocity scale is U_∞, the freestream velocity.

The particular flow studied in this paper is turbulent zero pressure gradient (ZPG) boundary layer flow. A pictorial representation of this flow is shown in Figure 1. While a full derivation of the theory is not pursued for the sake of brevity, a brief dimensional analysis is pursued below to elucidate the complicated behavior of transport in turbulent boundary layers. A complete derivation is given in the references [6], [7].

2.1 Momentum Transport in Turbulent Boundary Layer Flow

Momentum transport is dictated by the velocity field. The governing equation for the velocity field is given by the full Navier-Stokes equation:

$$\frac{\partial \tilde{u}_i}{\partial t} + \tilde{u}_j \frac{\partial \tilde{u}_i}{\partial x_j} = -\frac{1}{\rho}\frac{\partial \tilde{p}_i}{\partial x_i} + \nu \frac{\partial^2 \tilde{u}_i}{\partial x_j^2}, \tag{8}$$

where we assume there are no external body forces acting on the fluid. Einstein notation is used to indicate sums over all possible spatial directions. As is customary for turbulent flows, it is helpful to decompose the instantaneous velocity at specific position and time $\tilde{u}$ into a mean velocity U and fluctuating value u. This is called the Reynolds decomposition:

$$\tilde{u}(x, y, t) = U(x, y) + u(x, y, t), \tag{9}$$

where the following conditions on time-averaging must be satisfied:

$$\bar{u} = 0, \qquad \bar{\tilde{u}} = U = \lim_{T\to\infty} \frac{1}{T}\int_{t_0}^{t_0+T} \tilde{u}\, dt. \tag{10}$$

Assuming a ZPG, substituting the decomposition into equation (8) gives the following equation for momentum flow in the streamwise (x) direction:

$$\begin{aligned}\frac{\partial u_x}{\partial t} &+ (U_x + u_x)\frac{\partial (U_x + u_x)}{\partial x} \\ &+ (U_y + u_y)\frac{\partial (U_x + u_x)}{\partial y} = \nu \frac{\partial^2 (U_x + u_x)}{\partial x^2} + \nu \frac{\partial^2 (U_x + u_x)}{\partial y^2}.\end{aligned} \tag{11}$$

This simplifies to:

$$\frac{\partial u_x}{\partial t} + U_x \frac{\partial U_x}{\partial x} + U_y \frac{\partial U_x}{\partial y} + \frac{\partial (U_x u_x)}{\partial x} + \frac{\partial (U_y u_x)}{\partial y}$$
$$+ u_x \frac{\partial u_x}{\partial x} + u_y \frac{\partial u_x}{\partial y} = \nu \frac{\partial^2 (U_x + u_x)}{\partial x^2} + \nu \frac{\partial^2 (U_x + u_x)}{\partial y^2}. \tag{12}$$

Taking the ensemble average of equation (12) in order to get the average statistical behavior eliminates all the fluctuating terms in time and position. This gives:

$$U_x \frac{\partial U_x}{\partial x} + U_y \frac{\partial U_x}{\partial y} + \frac{\partial \langle u_x^2 \rangle}{\partial x} + \frac{\partial \langle u_x u_y \rangle}{\partial y} = \nu \frac{\partial^2 U_x}{\partial x^2} + \nu \frac{\partial^2 U_x}{\partial y^2}. \tag{13}$$

Using an order-of-magnitude analysis (see [5] for detailed derivation), the leading viscous terms with ν and the quadratic term in u_x become negligible with respect to the other momentum transport terms. Therefore, the governing equation for momentum transport in the streamwise direction becomes:

$$U_x \frac{\partial U_x}{\partial x} + U_y \frac{\partial U_x}{\partial y} = -\frac{\partial \langle u_x u_y \rangle}{\partial y}. \tag{14}$$

This is troublesome because the viscous terms have disappeared. The no-slip boundary condition cannot be satisfied without the presence of the viscous term. Therefore, it is evident that there must exist an inner boundary layer extremely close to the wall within the turbulent boundary layer where the viscous effects become important again.

Because of the existence of the inner boundary layer, there must exist 2 different sets of scaling parameters in the boundary layer. In the outer boundary layer, the velocity is nominally scaled by the free-stream velocity U_∞ and the length is scaled by the boundary layer thickness δ_{99}. But extremely near the wall, in the inner boundary layer, the characteristic velocity should not depend on U_∞. Therefore, a different velocity scale, u_τ, is chosen (it is referred to as friction velocity). Likewise, the length scale near the wall should not depend on δ_{99}. So an inner length scale η is chosen as well.

With these new scaling parameters applied to equation (13), the inner scale momentum equation may be obtained (see [5] for detailed derivation):

$$\frac{\partial}{\partial y} \left(\nu \frac{\partial U_x}{\partial y} - \langle u_x u_y \rangle \right) = 0. \tag{15}$$

Integrating equation (15) over y gives:

$$\nu \frac{\partial U_x}{\partial y} - \langle u_x u_y \rangle = \text{constant}. \tag{16}$$

Using the no-slip and kinematic boundary conditions, the velocity at the wall is 0, so $\langle u_x u_y \rangle|_{y=0} = 0$. Thus, constant $= \nu \left.\frac{\partial U_x}{\partial y}\right|_{y=0}$. The wall shear stress τ_w is defined as $\tau_w \equiv \mu \left.\frac{\partial U_x}{\partial y}\right|_{y=0}$; substituting this into the constant simplifies equation (16) to:

$$\nu \frac{\partial U_x}{\partial y} - \langle u_x u_y \rangle = \frac{\tau_w}{\rho}. \tag{17}$$

Then, non-dimensionalizing equation (17) in the usual way and applying the method of similarity solutions shows:

$$u_\tau = \sqrt{\frac{\tau_w}{\rho}}, \qquad \eta = \frac{\nu}{u_\tau}. \tag{18}$$

The inner boundary layer extends from $y = 0$ to $y = 0.1\delta_{99}$, while the outer boundary layer extends from $y = 100\eta$ to $y = \delta_{99}$. In many flows, $0.1\delta_{99} > 100\eta$, so there exists an overlap region in the boundary layer that is governed by both the inner and outer scaling parameters. The existence of this overlap allows the functional form of the velocity field to be constrained.

2.2 Heat Transport in Turbulent Boundary Layer Flow

A key facet of transport theory is the analogy between mass, momentum, and heat transport. All three types of transport are governed by similar conservation laws: Fick's Law of Diffusion for mass transport, Newton's Law of Viscosity for momentum transport, and Fourier's Law of Heating for heat transport [3].

The Reynolds analogy makes explicit the connection between turbulent momentum and heat transport. It states to first order that [4]:

$$\frac{H_y}{c_p \tau_{xy}} = -\frac{\partial \Theta / \partial y}{\partial U_x / \partial y}, \tag{19}$$

where the turbulent heat flux is given by $H_y = \rho c_p \langle u_y \theta \rangle$ and apparent turbulent momentum transfer is given by $\tau_{xy} = \rho \langle u_x u_y \rangle$. Θ denotes the mean temperature. Through this analogy, it is evident that parameters relating turbulent heat transfer and turbulent mo-

mentum transfer are directly related. Thus we can expect that the physics of turbulent heat transport will be similar to the physics of turbulent momentum transport.

Furthermore, in this paper, temperature is treated as a passive scalar quantity, meaning that temperature differences in the flow are small enough that they do not affect properties of the flow field itself [6]. In particular, this means that we only consider the situation where heat is transported by the velocity field and does not influence the momentum transport.

The governing equation for heat transport in a flow is the energy equation, which comes from the conservation of energy and Fourier's Law of Heating [8]:

$$\rho c_p \frac{\partial \tilde{\theta}}{\partial t} + \rho c_p \tilde{u_j} \frac{\partial \tilde{\theta}}{\partial x_j} = \tau_{ij} \frac{\partial \tilde{u_i}}{\partial x_j} + k \frac{\partial^2 \tilde{\theta}}{\partial x_j^2}, \tag{20}$$

where $\tilde{\theta}$ is the instantaneous temperature, $\tilde{u}$ is the instantaneous velocity, ρ is the fluid density (assumed constant), c_p is the specific heat capacity, τ_{ij} is the viscous shear stress tensor, and k is the thermal conductivity. Neglecting viscous shear stress due to the passive scalar nature of temperature and dividing by ρc_p allows equation (20) to be rewritten in the more suggestive form found in [4]:

$$\frac{\partial \tilde{\theta}}{\partial t} + \tilde{u_j} \frac{\partial \tilde{\theta}}{\partial x_j} = \gamma \frac{\partial^2 \tilde{\theta}}{\partial x_j^2}. \tag{21}$$

Here $\gamma = \dfrac{k}{\rho c_p}$ is the thermal diffusivity, which is a measure of how readily the medium diffuses heat. Non-dimensionalization in the usual way gives the following dimensionless constant, called the Prandtl number (Pr):

$$\mathrm{Pr} = \frac{\nu}{\gamma}, \tag{22}$$

where ν acts as measure of momentum diffusivity. In other words, the Prandtl number informs the relative thickness of the momentum boundary layer versus the thermal boundary layer. For air and most other gases, $\mathrm{Pr} \sim 1$, while for water $\mathrm{Pr} \sim 7$.

Assuming the usual flow setup given by Figure 1, the derivation of heat transport behavior proceeds almost identically to that of momentum transport. Using the Reynolds decomposition, the instantaneous temperature is decomposed into:

$$\tilde{\theta} = \Theta(x, y) + \theta(x, y, t), \tag{23}$$

where Θ is the mean temperature value and θ is the fluctuating temperature value. Plugging these decompositions into governing equation (21) and taking the ensemble average in the same manner as equation (13) gives:

$$U_x \frac{\partial \Theta}{\partial x} + U_y \frac{\partial \Theta}{\partial y} + \frac{\partial \langle u_x \theta \rangle}{\partial x} + \frac{\partial \langle u_y \theta \rangle}{\partial y} = \gamma \frac{\partial^2 \Theta}{\partial x^2} + \gamma \frac{\partial^2 \Theta}{\partial y^2}. \tag{24}$$

Using order of magnitude analysis again, the outer boundary layer heat diffusion equation is:

$$U_x \frac{\partial \Theta}{\partial x} + U_y \frac{\partial \Theta}{\partial y} = -\frac{\partial \langle u_y \theta \rangle}{\partial y}, \tag{25}$$

and the inner boundary layer heat diffusion equation is:

$$\frac{\partial}{\partial y} \left(\gamma \frac{\partial \Theta}{\partial y} - \langle u_y \theta \rangle \right) = 0. \tag{26}$$

These equations are identical in form to equations (14) and (15), as expected. Θ corresponds to U_x, θ corresponds to u_x, and γ corresponds to ν.

2.3 Thermal Boundary Layer Scaling Parameters

Although the physics governing turbulent momentum and heat transfer is similar, the temperature profiles are harder to solve because of Prandtl number dependence, which introduces more parameters to the governing equations. There exist multiple proposed theories for the mean temperature profile Θ, temperature variance profile $\langle \theta^2 \rangle$, and their scalings. In particular, the temperature variance profile at $\mathrm{Pr} > 1$ and the validity of its proposed scalings has not been explored thoroughly in the existing literature.

Furthermore, because the Prandtl number of water is approximately 7, the thermal boundary layer is approximately 2 times thinner than the velocity boundary layer, since the thickness goes like $\mathrm{Pr}^{-1/3}$. Boundary layers for high Reynolds number flows are already extremely thin and difficult to measure, so confirming the inner thermal boundary layer scalings at higher Prandtl numbers is particularly difficult since the smallest scale turbulent flow features to be measured are on the order of microns in length.

With this in mind, two proposed mean temperature scalings are given below. The first is due to Byers [6] and the second is due to Kader [9].

According to Byers, the inner equation can be scaled by expressing the mean tem-

perature relative to a fixed temperature, and dividing that quantity by the scaling parameter Θ_τ. The most obvious choice for the fixed temperature in the inner boundary layer is the wall temperature Θ_w. The resulting dimensionless temperature Θ^+ will be a function of non-dimensional y-coordinate $y^+ \equiv \dfrac{y}{\eta}$ and Prandtl number. Hence:

$$\Theta^+ \equiv \frac{\Theta - \Theta_w}{\Theta_\tau} = f_i(y^+, \mathrm{Pr}). \tag{27}$$

Integrating equation (26) over y and using the boundary condition at $y = 0$ yields:

$$\frac{q_w}{\rho c_p} = \gamma \frac{\partial \Theta}{\partial y} - \langle u_y \theta \rangle, \tag{28}$$

where q_w denotes the wall heat flux. Substituting equation (27) into (28) and using the relevant scalings for y and u_y gives:

$$1 = \frac{\gamma \rho c_p \Theta_\tau}{q_w \eta_T} \frac{\partial f}{\partial y^+} + \frac{\rho c_p H_c}{q_w} h, \tag{29}$$

where turbulent heat flux $\langle u_y \theta \rangle$ was scaled with H_c, yielding non-dimensional function h. Since all terms in this non-dimensional equation are of order 1, it can be concluded that the scaling parameter for the inner temperature scale is:

$$\Theta_\tau = \frac{q_w \eta_T}{\gamma \rho c_p} = \frac{q_w}{\rho c_p u_\tau}. \tag{30}$$

The outer equation is scaled in a nearly identical manner, except that temperature is defined relative to freestream temperature Θ_∞, and then is divided by scaling parameter Θ_o. Therefore:

$$\frac{\Theta - \Theta_\infty}{\Theta_o} = f_o(\bar{y}, \mathrm{Pr}), \tag{31}$$

where f_o is a non-dimensional function of dimensionless distance $\bar{y} \equiv y/\delta_{99}$ and Prandtl number. After a similar non-dimensionalization procedure (see [6] for full derivation), it is found that

$$\Theta_o = \frac{q_w}{\rho c_p u_\tau} \frac{U_\infty}{u_\tau} = \Theta_\tau \frac{U_\infty}{u_\tau}. \tag{32}$$

Byers notes that choosing different temperature and length scales will result in different scalings of the inner and outer boundary layers. The scalings found in equation (30) and (32) are difficult to confirm experimentally because they depend on the wall heat flux, which can be difficult to measure. Instead, in a seminal paper on temperature

profiles in turbulent boundary layers, Kader directly calculates Θ^+, which provides an indirect way of calculating Θ_τ in terms of measurable quantities.

Kader proposes that the dimensionless temperature Θ^+ must satisfy the following governing equation [9], which is ultimately derived from the energy equation:

$$\frac{d\Theta^+}{dy^+} = (\mathrm{Pr}^{-1} + \epsilon/\nu), \quad \Theta^+(0) = 0, \quad \Theta^+(\delta_{99}) = \Theta_o^+, \tag{33}$$

where ϵ is the eddy diffusivity of heat (a measure of effectiveness of turbulent heat transfer) and Θ_o^+ is the dimensionless temperature at the free stream. Kader defines the dimensionless temperature as:

$$\Theta^+ = \frac{\Theta_w - \Theta}{\Theta_\tau}, \tag{34}$$

where Θ_w is the temperature at the wall and Θ_τ is the scaling parameter, which Kader defines to be the same as equation (30).

Using physical intuition and existing experimental data, several conditions must be satisfied by Θ^+ in order for it to be considered a good approximation to the universal behavior of the mean temperature field of a turbulent boundary layer [9]. They are listed below.

1. Very near the wall in the inner boundary layer, Θ^+ is linear in y^+:

   $$\Theta^+ = \mathrm{Pr} \cdot y^+. \tag{35}$$

   This arises from the fact that in the inner boundary layer, molecular heat transfer from the wall should dominate over turbulent heat transfer. Therefore, $\epsilon = 0$; substituting this condition into equation (33) enables us to obtain equation (35).

2. For moderate Prandtl numbers (i.e. $\mathrm{Pr} \sim 1$), the thickness of the inner boundary layer is $y^+ \cong 30$, where this y^+ denotes the number of η units from the wall to the edge of the inner boundary layer.

3. To higher order, Θ^+ deviates from its linear behavior near the wall, with deviations beginning at $y^+ \sim 9/\mathrm{Pr}^{1/3}$.

4. Further away from the wall, the dimensionless temperature is logarithmic and given by:

   $$\Theta^+ = \alpha \ln y^+ + \beta(\mathrm{Pr}), \tag{36}$$

   where $\beta(\mathrm{Pr})$ is the temperature difference between the wall and the lower edge of

the outer boundary layer, and is calculated experimentally to be:

$$\beta(\mathrm{Pr}) = (3.85 \cdot \mathrm{Pr}^{1/3} - 1.3)^2 + 2.12 \ln \mathrm{Pr}, \tag{37}$$

holding for Prandtl numbers from 0.006 to 40000.

5. The molecular properties of the fluid do not influence Θ^+, and when the free-stream is reached, Θ^+ does not change anymore with respect to y^+.

Kader proposed that all of these conditions are satisfied by the following equation for Θ^+, which he derived using a numerical interpolation formula:

$$\Theta^+ = \mathrm{Pr} \cdot y^+ e^{-\Gamma} + \left[2.12 \ln \left[(1 + y^+) \frac{2.5(2 - y/\delta_{99})}{1 + 4(1 - y/\delta_{99})^2} \right] + \beta(\mathrm{Pr}) \right] e^{-1/\Gamma}, \tag{38}$$

where

$$\Gamma = \frac{0.01(\mathrm{Pr} \cdot y^+)^4}{1 + 5\mathrm{Pr}^3 \cdot y^+}. \tag{39}$$

Because Pr, y^+, and δ_{99} can all be measured, equation (38) provides a closed-form way of finding Θ^+. More importantly, since Θ_w and Θ are measurable, equations (34) and (38) provide a way to obtain Θ_τ. This scaling will henceforth be referred to as the Kader scaling.

According to Byers, similarity theory dictates that finding the correct scaling parameters for the mean temperature field Θ will allow us to scale the temperature variance field $\langle \theta^2 \rangle$ in a very similar manner [6]. More specifically, if a similarity solution exists for the temperature field, it must scale both the mean and the variance. In the case of momentum transport, for example, Hultmark found that for the velocity field, the inner and outer variance scaling parameters were equivalent to the squares of the inner and outer mean scaling parameters [10]. Byers thus proposed that the scaling parameters for the temperature variance are also the squares of the scaling parameters for the mean temperature.

Therefore in this paper, we experimentally test the proposed theoretical scaling parameters for the mean and variance temperature field at moderately high Prandtl number of Pr = 6.6. We use the Kader scaling to scale the mean temperature measurements with Θ_τ; if the collapse of mean data is reasonable, we apply the similarity scaling to the temperature variance data and assess the collapse of the variance data.

3 Experimental Setup and Data Collection

The experiment was conducted at the Gas Dynamics Laboratory on the Forrestal Campus of Princeton University. The experimental setup consisted of a large, re-circulating water tunnel through which water flowed. The conditions of the tunnel were manipulated in order to produce a turbulent boundary layer flow over a flat wall. Measurements of the temperature and velocity of the flow field inside the boundary layer were taken with a new measuring device developed by the Hultmark lab of Princeton University.

3.1 The Nanoscale Cold-Wire Resistance Thermometer Probe

The instrument used to take measurements of the temperature field was a modified nanoscale cold-wire resistance thermometer probe called FT-NSTAP. The sensor in the probe consists of a nanoscale piece of platinum whose cold resistance, which can be measured, is linearly correlated to the local temperature of the fluid in contact with the probe [11]. Furthermore, the probe was designed to mitigate end conduction effects that dampen turbulent fluctuations and may cause errant measurements [12].

Due to stresses exerted by water proportional to its density, the FT-NSTAP was made to be robust to ensure it did not deform in the water while taking measurements. The platinum ribbon in the sensor has dimensions of 30 μm length $\times$ 2μm width $\times$ 0.5 μm thickness, making it 5 times thicker than a typical probe [11]. However, the small length of the probe greatly increases the spatial resolution of the sensor, which allowed the probe to make accurate measurements of the smallest turbulent features. The probe also has a temporal bandwidth of approximately 1 MHz in water. An image of the probe is shown in Figure 2a.

3.2 The Re-circulating Water Tunnel and Heating Elements

The boundary layer measurements were obtained from a recirculating water tunnel with test section dimensions of 1 m $\times$ 0.15 m $\times$ 0.2 m, a 3600 L reservoir, and water flow speeds between 0.1 m/s and 10 m/s. The water itself was deionized to reduce electrical noise in the sensor's signal. In order to facilitate the development of a turbulent boundary layer, the flow was tripped at the leading edge of the test section with a triangular grid of thickness 0.45 mm. Tripping induces turbulence before the natural laminar-to-turbulent

flow transition by introducing a large disturbance in the boundary layer [1]. An image of the water tunnel setup is shown in Figure 2b.

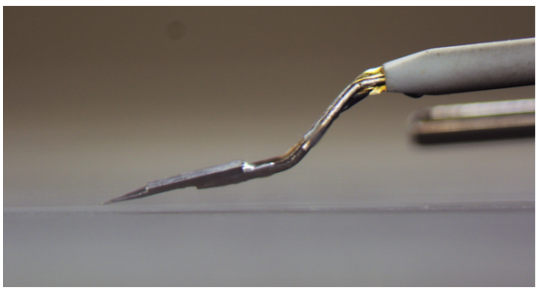
Heating was induced in the boundary layer using 42 1-kW heaters controlled by 14 temperature controllers embedded just below the floor of the tunnel. This allowed an approximately constant wall temperature along the floor of the test section to be maintained, at a value of 5 degrees Kelvin above the water temperature. The wall temperature was set to be 301 degrees Kelvin (28 degrees Celsius), and the free-stream temperature was measured to be 296 degrees Kelvin (23 degrees Celsius). Boundary layer measurements of the temperature field were made on the centerline of the test section approximately 750 mm downstream from the trip. The sensor itself moved up through the boundary layer using a linear traverse and encoder in 0.5 μm increments, allowing it to take 40 logarithmically spaced measurements starting at the wall and moving up to the free-stream.

Experiments were carried out on the water tunnel by independently varying the speed of water flow in the tunnel. This changed the Re_τ number, called the friction Reynolds number, and altered the dynamics of each run. The friction Reynolds number is defined as:

$$\mathrm{Re}_\tau \equiv \frac{\delta_{99} u_\tau}{\nu}, \tag{40}$$

where δ_{99} is the boundary layer thickness, u_τ is the friction velocity as defined from equation (18), and ν is the kinematic viscosity. All runs were carried out at the same Prandtl number of $\mathrm{Pr} = 6.6$.

15 runs were conducted, with Re_τ numbers distributed approximately from $\mathrm{Re}_\tau = 500$ to $\mathrm{Re}_\tau = 1200$. In each run, the FT-NSTAP was first used as a hotwire anemometer in conjunction with a Pitot tube to measure the following information about the flow: δ_{99} value, η value, and u_τ value. Then, the probe was switched to its default coldwire setting to measure the mean temperature Θ and temperature variance $\langle \theta^2 \rangle$ in the boundary layer.



(a) The FT-NSTAP being used to take measurements of the thermal boundary layer very near the wall. Photo credit: Matt Fu.

(b) Setup of the re-circulating water tunnel. Measurements were taken in the clear rectangular region in the center, as water flowed from left to right. The trip was located inside the metal box to the left of the clear rectangle. Heaters were controlled by the metal box in the bottom left corner. Photo credit: Marcel Louis.

Figure 2: Measuring device and experimental setup.

4 Results

Using the data gathered from the experiment, Θ_τ was calculated using the Kader scaling. According to equation (34), Θ_τ is given by the formula:

$$\Theta_\tau = \frac{\Theta_w - \Theta(y)}{\Theta^+(y)}. \tag{41}$$

In particular, (41) must hold at the boundary between the outer boundary layer and the free-stream. Thus, we have:

$$\Theta_\tau = \frac{\Theta_w - \Theta_\infty}{\Theta^+(y \approx \delta_{99})}, \tag{42}$$

where Θ_w is the wall temperature, Θ_∞ is the free-stream temperature, and $\Theta^+(y)$ is given by equation (38). Because of unavoidable experimental error, two different methods were used to calculate $\Theta_w - \Theta_\infty$.

The first method used to calculate Θ_τ assumed that the difference between wall

temperature and free-stream temperature was always 5 degrees Kelvin. The experiment was set up so that the initial conditions did satisfy this condition. However, the temperature controllers only report the temperature read at their sensor embedded just below the wall. The controllers had no way of telling what the actual wall temperature was in real-time, which could have easily deviated slightly from its imposed value of 301 K due to inadvertent heat flux from the outside environment or the fluid itself.

The second method used to calculate Θ_τ measured the quantity $\Theta_w - \Theta_\infty$ by taking the difference in mean temperature values of the first and last measurement of each run. This method assumed that the first measurement was so close to the wall that it effectively measured the wall temperature, whereas the last measurement was nearly always measured in the free-stream and thus was the free-stream temperature value.

Both methods were applied to the mean temperature and temperature variance measurements of all 15 runs. It was found that in every single case except one, the second method resulted in better collapse of the data. The first method yielded a better collapse in the data when temperature was scaled from the free-stream. The Θ_τ-scaled results are shown below. The legend in each figure denotes the different Re_τ numbers that correspond to different runs.

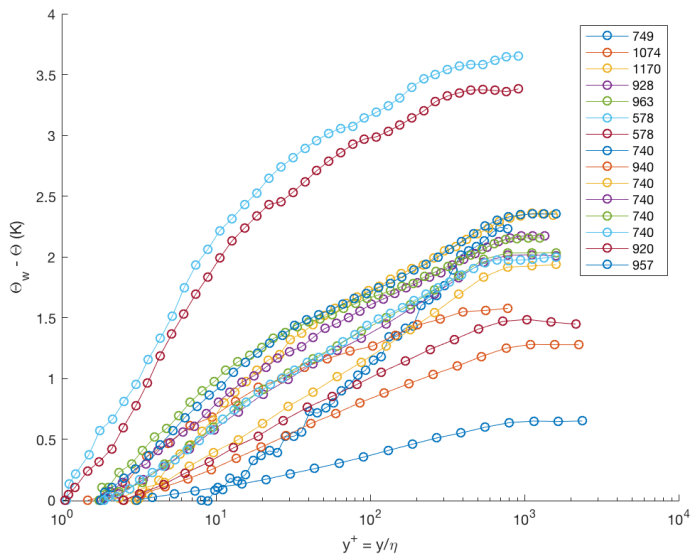


Figure 3: Raw measurements of the mean temperature Θ measured relative to the wall temperature Θ_w. y-distance is normalized with inner scaling η and plotted logarithmically.

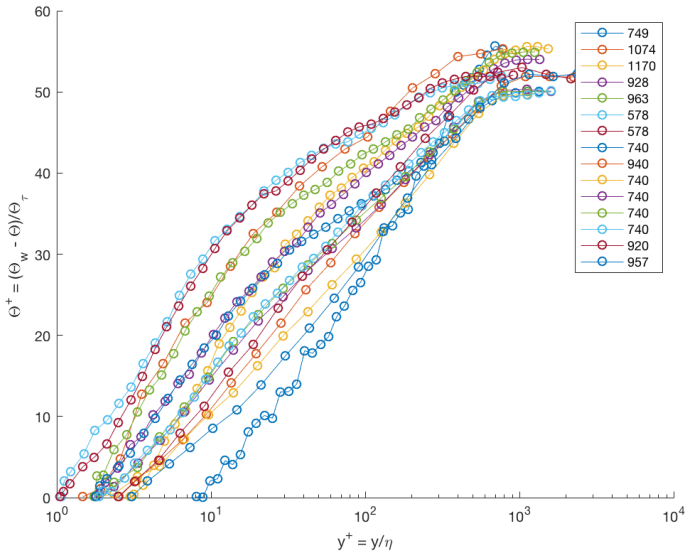


Figure 4: Mean temperature profile scaled using Θ_τ calculated with the second method. y-distance is normalized with inner scaling η and plotted logarithmically.

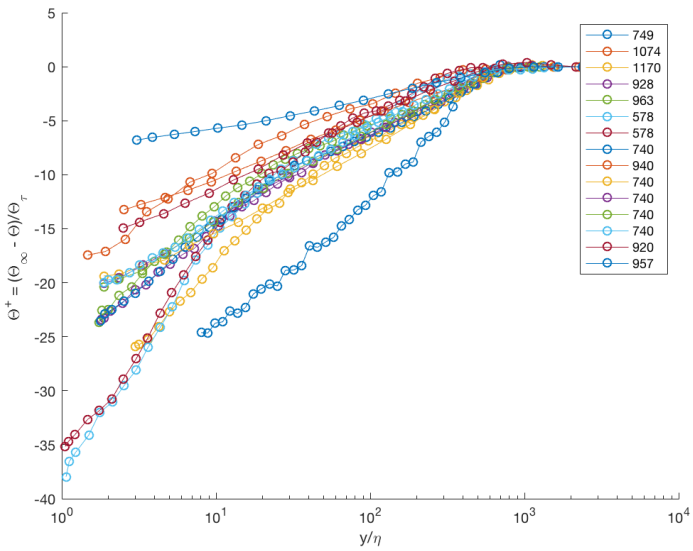


Figure 5: Mean temperature profile scaled using Θ_τ calculated with the first method, and measured relative to the free-stream temperature Θ_∞. y-distance is normalized with inner scaling η and plotted logarithmically.

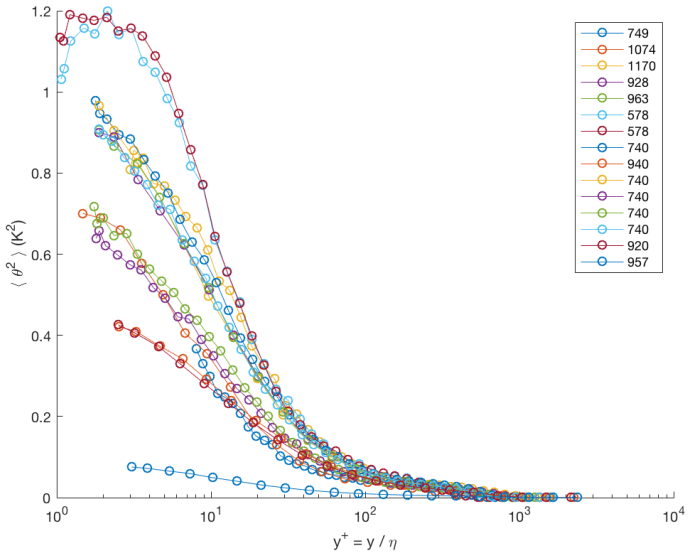


Figure 6: Raw measurements of the temperature variance $\langle \theta^2 \rangle$. y-distance is normalized with inner scaling η and plotted logarithmically.

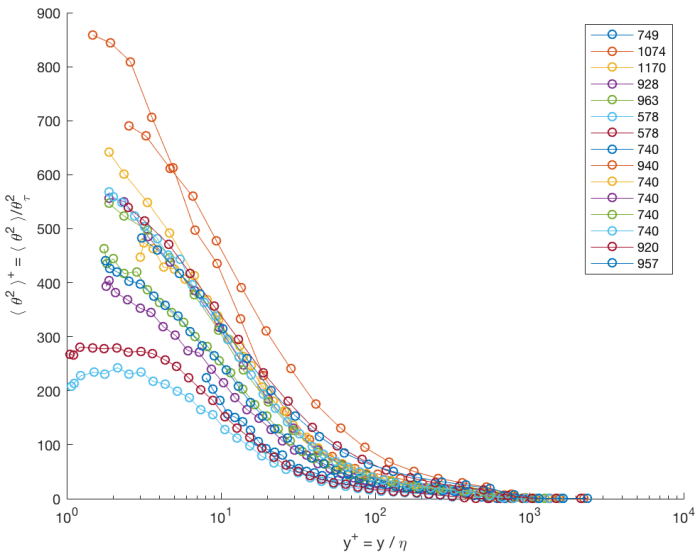


Figure 7: Temperature variance profile scaled using Θ_τ^2 calculated with the second method. y-distance is normalized with inner scaling η and plotted logarithmically.

Due to the stochastic behavior of turbulence, it is difficult to obtain a meaningful

quantitative measure of the range of error in the data obtained on the mean and variance temperatures. In each run, the sensor was run for long enough to obtain convergent statistics consisting of thousands of statistically independent measurements. Hence it can be reasonably assumed that the errors in the statistical estimators should be within a few percent. Therefore the collapse of the data will be analyzed directly from the figures. A rigorous error analysis requires the experimental errors to be separated from the stochastic variance of the data, and can be conducted in the future once more data is obtained from the experiment. Possible sources of experimental error include the uncertainty in the wall temperature (as discussed earlier in this section) as well as measurement errors from the cold-wire probe.

5 Conclusions and Future Work

The preliminary data obtained from this experiment shows that the Kader scaling performs reasonably well in collapsing the mean and variance temperature fields from their raw values in Figures 3 and 6 to the scaled values in Figures 4, 5, and 7. According to Figures 4 and 7, the collapse of the mean and variance temperature profiles due to the second method of calculating Θ_τ is particularly good at $y^+ > 100$, which usually corresponds to the outer boundary layer region. For $10 < y^+ < 100$, an increase in the spread between different runs becomes noticeable, indicating that the scaling is not as good in collapsing the data in that region. For $y^+ < 10$, there significant discrepancies appear in where the mean temperature profile reaches the wall. Further, the expected linear behavior in the mean temperature profile and the peak in the variance profile very near the wall at $y^+ < 10$ did not appear. Both the spread and these discrepancies must be resolved through an improvement in measurement of the temperature field very near the wall. Note that $y^+ < 10$ usually corresponds to distances of less than 100 microns away from the wall.

Among the various Θ profiles, the best collapse was actually found when temperature was measured relative to the free-stream temperature and not the wall temperature, as shown in Figure 5. In this particular case, Θ_τ was calculated using the first method. This suggests that the current combination of parameters tested is not necessarily the optimal combination for collapsing the temperature profiles, and that there most likely exist other scalings that may better collapse the data.

In the future, more proposed scalings in the literature will be tested to assess which

scaling produces the best collapse of the mean and variance temperature profiles. For example, instead of using the Kader scaling to indirectly obtain Θ_τ through calculating Θ^+, a method of directly measuring the wall heat flux q_w will be pursued. This may produce a better estimate of Θ_τ using the Byers scaling since it would allow us to directly calculate the scaling parameter. Doing a polynomial fit of the data using Kader's scaling instead of only using the free-stream and wall temperature values to calculate Θ_τ can also be pursued to provide a better picture of the validity of the Kader scaling.

Overall, we conclude the Kader scaling is reasonable effective in the outer region of the boundary layer, with its effectiveness decreasing as we move closer to the wall. This furthers our knowledge of how to parameterize heat transfer in a turbulent boundary layer in order to find a similarity solution for the flow.

A Appendix: Similarity Solution Method

The goal of using similarity solutions is to reduce the complexity of partial differential equations. This is achieved by using a coordinate transformation that combines several of the existing independent variables and recasts the equation in terms of the new "similarity" coordinate. Dimensional analysis and correct choice of scaling parameters helps inform what the similarity coordinate should be. To illustrate this approach, an example using the similarity solution method to derive the velocity field of a zero-pressure gradient laminar boundary layer flow is given below [13].

The governing equations for flow in a ZPG laminar boundary layer are:

$$u_x \frac{\partial u_x}{\partial x} + u_y \frac{\partial u_x}{\partial y} = \nu \frac{\partial^2 u_x}{\partial y^2}, \tag{43}$$

and

$$\frac{\partial u_x}{\partial x} + \frac{\partial u_y}{\partial y} = 0. \tag{44}$$

The goal is to solve for the universal behavior of the velocity field $\vec{u} = (u_x, u_y)$. Using dimensional analysis, we propose a non-dimensional similarity solution variable called:

$$\eta(x, y) \equiv \frac{y}{\delta} = y\sqrt{\frac{u_\infty}{\nu x}}. \tag{45}$$

Now, we reduce our system of coupled PDEs to one equation by introducing the stream function $\Psi(x, y)$ such that:

$$u_x \equiv \frac{\partial \Psi}{\partial y}, \qquad u_y \equiv -\frac{\partial \Psi}{\partial x}. \tag{46}$$

Substituting these definitions into equation (44) shows that the continuity equation is still satisfied. Next, we non-dimensionalize u_x and Ψ, so that:

$$\Psi = \Psi_0 f(\eta), \qquad u_x = u_\infty g(\eta). \tag{47}$$

Plugging in Ψ into equation (46) gives:

$$u_x = \frac{\partial}{\partial y}(\Psi_0 f(\eta)) = \frac{\partial}{\partial \eta}(\Psi_0 f(\eta))\frac{\partial \eta}{\partial y} \tag{48}$$

$$= \Psi_0 \frac{\partial \eta}{\partial y}\frac{df(\eta)}{d\eta} = \Psi_0 \sqrt{\frac{\infty}{\nu x}} f'(\eta) \tag{49}$$

$$\implies f'(\eta) = \frac{u_x \delta}{\Psi_0}. \tag{50}$$

By the rules of non-dimensionalization, we are free to choose what Ψ_0 can be. Therefore, we conveniently choose Ψ_0 such that $\frac{\Psi_0}{\delta} = u_\infty$. This makes $f'(\eta) = \frac{u_x}{u_\infty} = g(\eta)$, which thus reduces functional dependency to one variable, η. Hence:

$$\Psi_0 = \delta u_\infty = u_\infty \sqrt{\frac{\nu x}{u_\infty}} = \sqrt{\nu x u_\infty}. \tag{51}$$

Now that we know one of our normalization constants, Ψ_0, we find u_y:

$$u_y = -\frac{\partial \Psi}{\partial x} = -\frac{\partial}{\partial x}(\Psi_0 f(\eta)) = -\frac{\partial}{\partial x}(\sqrt{\nu x u_\infty} f(\eta)) \tag{52}$$

$$= -\left(f \cdot \frac{\partial}{\partial x}(\sqrt{\nu x u_\infty}) + \sqrt{\nu x u_\infty}\frac{\partial f}{\partial \eta}\frac{\partial \eta}{\partial x}\right) \tag{53}$$

$$= -\left(\frac{f}{2}\sqrt{\frac{\nu u_\infty}{x}} - \frac{y}{2x} u_\infty f'(\eta)\right) \tag{54}$$

$$\implies u_y = \sqrt{\frac{\nu u_\infty}{x}}\frac{(\eta f'(\eta) - f(\eta))}{2}. \tag{55}$$

Substituting u_x and u_y into the governing Navier-Stokes equation (equation (43)) gives, after simplification:

$$-\frac{u_\infty}{2x} f' f'' \eta + \frac{u_\infty}{2x}(\eta f' - f) f'' = \frac{u_\infty}{x} f''' \tag{56}$$

$$\implies -\frac{u_\infty}{2x} f \cdot f'' = \frac{u_\infty}{x} f'''. \tag{57}$$

Thus, the similarity solution method has reduced the initial set of 2 coupled PDEs into the following numerically solvable ODE:

$$\frac{d^3 f(\eta)}{d\eta^3} + \frac{1}{2}\frac{d^2 f(\eta)}{d\eta^2} \cdot f(\eta) = 0, \qquad \eta = y\sqrt{\frac{u_\infty}{\nu x}}. \tag{58}$$

Solving for $f(\eta)$ will in turn give u_x and u_y by equations (50) and (55), as desired.

References

[1] A.J. Smits. *A Physical Introduction to Fluid Mechanics.* John Wiley, 2018.

[2] C.K. Batchelor, G.K. Batchelor, and Cambridge University Press. *An Introduction to Fluid Dynamics.* Cambridge Mathematical Library. Cambridge University Press, 1967.

[3] F. Reif. *Fundamentals of Statistical and Thermal Physics.* Waveland Press, 2009.

[4] H. Tennekes and J.L. Lumley. *A First Course in Turbulence.* Pe Men Book Company, 1972.

[5] William K. George. Lectures in turbulence for the 21st century, 2013.

[6] Clayton P. Byers. *Theoretical and experimental investigations of similarity solutions in turbulent flows.* PhD thesis, Princeton University, 2018.

[7] William K. George and Luciano Castillo. Zero-pressure-gradient turbulent boundary layer. *Applied Mechanics Reviews*, 50:689–729, 1997.

[8] H. Schlichting and K. Gersten. *Boundary-Layer Theory.* Springer Berlin Heidelberg, 2003.

[9] B.A. Kader. Temperature and concentration profiles in fully turbulent boundary layers. *International Journal of Heat and Mass Transfer*, 24(9):1541–1544, 1981.

[10] Marcus Hultmark. A theory for the streamwise turbulent fluctuations in high reynolds number pipe flow. *Journal of Fluid Mechanics*, 707:575–584, 2012.

[11] Matthew K. Fu, Clayton P. Byers, Katherine A. Kokmanian, and Marcus N. Hultmark. Nanoscale cold-wire measurements of temperature fluctuations in a heated turbulent boundary layer. Manuscript submitted for publication, 2019.

[12] Gilad Arwatz, Carla Bahri, Alexander J Smits, and Marcus Hultmark. Dynamic calibration and modeling of a cold wire for temperature measurement. *Measurement Science and Technology*, 24(12):125301, 2013.

[13] L.D. Landau and E.M. Lifshitz. *Fluid Mechanics.* Elsevier Science, 2013.

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